

Xingyun Mou

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EDUCATION

University of Southern California

Los Angeles, CA

Candidate for Bachelor of Science in Applied and Computational Mathematics

Jan 2024 - Present

Minor in Artificial Intelligence Applications

- **Cumulative GPA: 3.83/4.0**
- **Honor:** 2024 Dornsife Dean's List, 2025 Dornsife Dean's List, 2026 Dornsife Dean's List
- **Relevant Courses:** Artificial Intelligence, Object Oriented Programming, Discrete Methods in Computer Science, Machine Learning, Probability Theory, Neural Network

EXPERIENCE

University of Southern California

Los Angeles, CA

Undergraduate Research Assistant

Feb 2026 – Present

- Conducted research on extracting object-level physical properties and structural information from motion using PSI world model and SpelkeNet
- Performed virtual poking experiments by specifying hold and movement constraints and analyzing the resulting sparse and dense optical flow fields
- Extended insights from video-based vibrometry to investigate whether PSI-predicted motion under virtual interactions can reveal object stiffness, flexibility, and physical structure

DSO Summer Scholars

Los Angeles, CA

Undergraduate Researcher

May 2026 – August 2026

- Developed a Python-based discrete-event simulation to model call center queueing, abandonment, redial behavior, and repeated caller attempts
- Built an interactive Streamlit dashboard to run parameter sweeps, visualize attempt distributions, and evaluate policy sensitivity
- Designed a Monte Carlo evaluation framework to compare operational performance across three call centers and multiple policies

Shanghai AI Laboratory

Shanghai, China

Research Intern

May 2025 - Dec 2025

- Fine-tuned DeepSeek-R1-Distill-Qwen-7B using Chain-of-Thought supervision to improve multi-step mathematical reasoning
- Built and preprocessed large-scale datasets for model pretraining, including data cleaning, normalization, and formatting to support stable training runs
- Investigated Lean4-based formal verification in improving LLM mathematical reasoning capabilities

Math Learning Assistant

Los Angeles, CA

Learning Assistant

Sep 2024 - Dec 2024

- Assisted students during weekly sessions by answering questions and emphasizing key solution steps
- Explained core Calculus II concepts using structured problem-solving approaches

University of California Irvine

Irvine, CA

Undergraduate Research Assistant

Nov 2022 - Nov 2023

- Processed and analyzed large-scale datasets (satellite imagery, geospatial layers, time-series data) to extract structured features for modeling
- Used QGIS and remote sensing data to perform 30-year land cover change analysis
- Identified nonlinear relationships and regional patterns in high-dimensional data using exploratory data analysis and correlation analysis

SKILLS AND INTERESTS

Language: Fluent - English and Native - Mandarin

Technical Skills: C++, Python, R, Git, Linux, MATLAB, QGIS, EXCEL, LaTeX